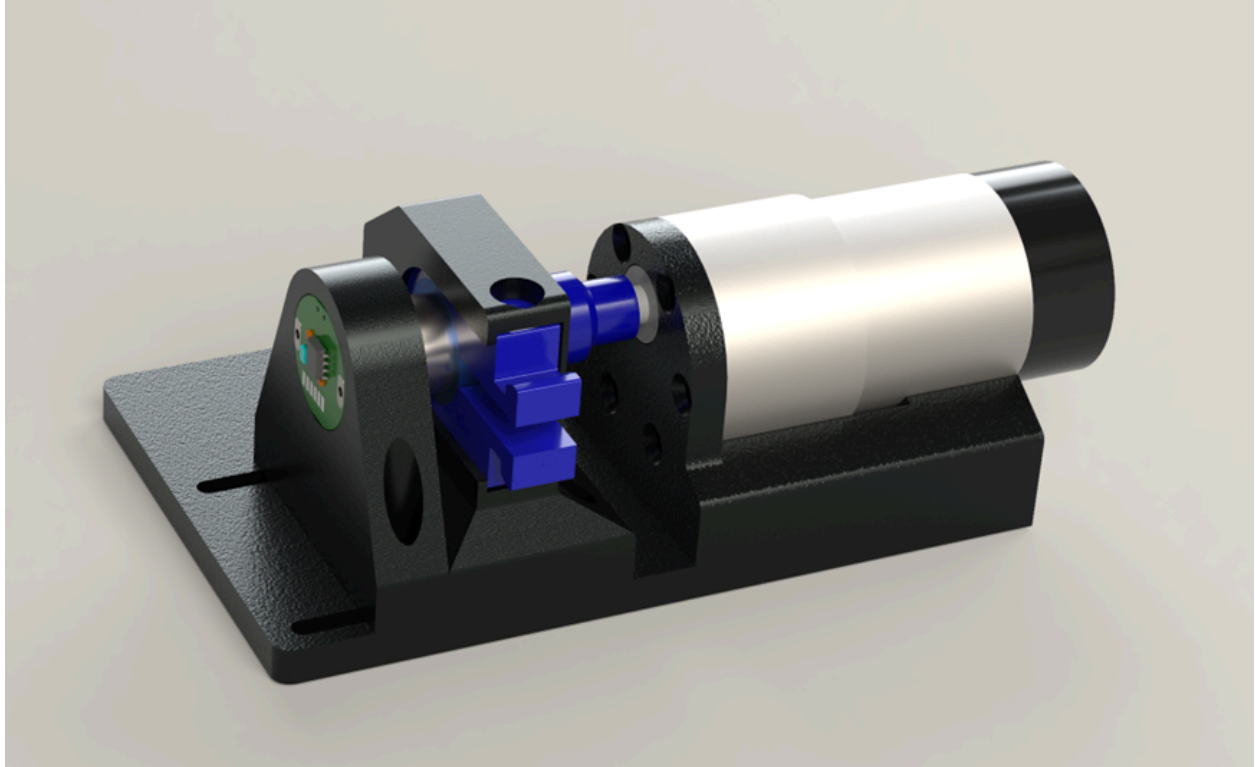




ES-259 Final Project

“Kinematic Modeling and Control of Backlash and Friction in Manipulators”

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Abstract:

Robotic manipulators often deviate from ideal behavior due to nonlinear effects such as friction and backlash, which introduce dead zones, hysteresis, and tracking error. This project presents a model-based control framework that explicitly identifies and compensates for these nonlinearities within the control loop. Friction is characterized using a Coulomb model, while backlash is captured as a dead-zone effect based on experimental system data. These models are incorporated into a controller that combines feedforward compensation with feedback control to improve tracking performance. The approach is evaluated on a system with incrementally adjustable levels of friction and backlash, enabling controlled testing under varying nonlinear conditions. Experimental results show that the proposed controller reduces tracking error by approximately 50% compared to a baseline controller, with consistent performance across increasing levels of nonlinearity. Controllers that explicitly account for system nonlinearities can significantly improve the accuracy and reliability of robotic manipulators, particularly in systems with brushed motors where friction and backlash are prominent.

1. Introduction:

A major challenge in controlling robotic manipulators arises from nonlinearities such as backlash and friction, which cause real systems to deviate from ideal models. This project presents a control framework that models these effects and compensates for them to improve system accuracy.

1.1 Project Goals

The primary goal of this project is to develop and validate a model that identifies and quantifies backlash and friction in a robotic joint, and to use these parameters to refine the control law and reduce system tracking error. Specifically, this project aims to:

- Develop a model to capture backlash as a dead-zone nonlinearity
- Identify friction parameters using a Coulomb friction model
- Implement a control framework combining feedforward compensation and feedback control
- Evaluate system performance through simulation and experimental validation

1.2 Problem Interest

In robotic systems, control design is often based on idealized models that neglect nonlinear effects such as friction and backlash. However, these nonlinearities are unavoidable in real-world systems and can significantly degrade accuracy and repeatability. This is particularly critical in high-precision applications, where even small deviations can lead to substantial performance errors, such as in robotic surgery, high-precision manufacturing, or manipulation tasks.

In our own robotics work, including the development of a six degree-of-freedom robotic arm for the Harvard Undergraduate Robotics Club, backlash has been a persistent challenge in achieving reliable control performance. These experiences motivated a deeper investigation into modeling and compensating for backlash and friction, rather than treating them as minor disturbances.

2. Background:

2.1 Backlash

Backlash is the gap between actuator motion and joint response, which results in a dead-zone when the direction of actuation changes. This phenomenon is commonly observed in geared systems (Fig. 1), where a finite amount of mechanical play exists due to limitations in machining precision, gear tooth spacing, and assembly tolerances. As a result, small changes in actuator position may not immediately produce motion at the output, leading to response delay and variation in behavior depending on actuation direction.

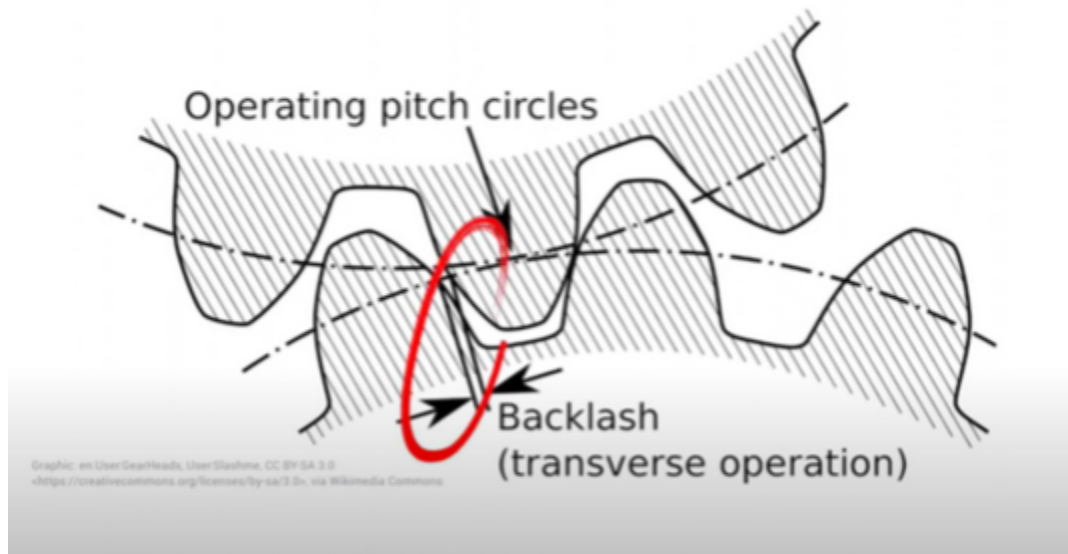
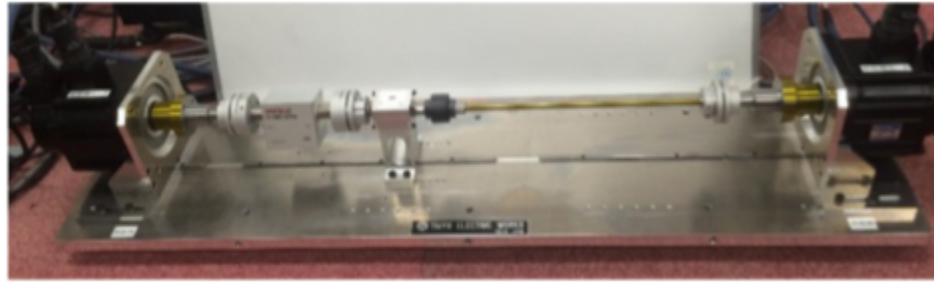


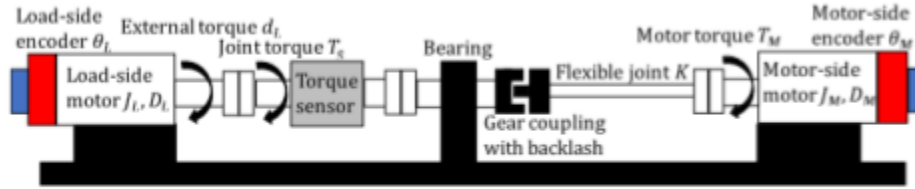
Figure 1: Backlash between two meshed gears [1]

In control-oriented modeling, backlash is often represented as a nonlinear dead-zone with hysteresis between the motor-side and load-side positions. A common framework for capturing this behavior is the two-inertia system, where the motor and load are treated as separate dynamic bodies connected through a nonlinear element representing the backlash gap. In this formulation, motion is transmitted only when the relative displacement between the motor and load exceeds the backlash threshold, producing discontinuous engagement and disengagement during operation.

Yamada and Fujimoto (2019) demonstrate that backlash significantly degrades tracking performance, particularly during direction reversals, where the system must traverse the dead-zone before torque is transmitted to the load [2]. Their work shows that accurate compensation requires knowledge of both motor-side and load-side states, motivating the use of dual encoders for precise measurement. By explicitly modeling the backlash gap and incorporating it into the control design, they achieve improved torque transmission and reduced tracking error. This highlights the importance of treating backlash as a structured nonlinearity rather than a minor disturbance in high-precision control applications. Additionally, their apparatus diagram served as inspiration for our own (Fig. 2)



(a) Outlook of the motor bench



(b) Schematic of the motor bench

Figure 2: Appartus setup from Yamada, S., & Fujimoto, H. (2019)

Márton and Lantos (2009) further investigate the combined effects of backlash and nonlinear friction in control systems, modeling backlash as an uncertain dead-zone within the system dynamics [3]. Their work emphasizes that backlash parameters are often unknown or difficult to measure directly, requiring robust or optimal control strategies, such as linear quadratic (LQ) control, to maintain stability and performance. LQ control is an optimal control method that determines the control input by minimizing a cost function that penalizes both tracking error and control effort. By balancing these competing objectives, the controller produces smooth and stable system behavior even in the presence of uncertainties. They show that the interaction between backlash and friction can lead to complex behaviors such as oscillations and limit cycles, reinforcing the need for control approaches that explicitly account for both nonlinearities.

2.2 Friction

Friction is one of the most significant nonlinearities affecting robotic systems, as it directly impacts tracking performance, stability, and energy efficiency. In robotic manipulators, friction arises at joints and contact interfaces and can vary depending on velocity, load, and surface conditions. Common friction components include Coulomb friction (constant opposing force), viscous friction (proportional to velocity), and velocity-dependent effects such as the Stribeck effect, which captures the reduction in friction at low velocities. These nonlinear behaviors are particularly prominent in systems operating under varying loads or low-speed conditions. In addition to friction at the joint level, friction is also present within the actuator itself (Fig. 3), particularly in brushed DC motors. Internal sources such as brush contact, bearing resistance, and gearbox friction contribute to the overall nonlinear behavior of the system. These effects are especially pronounced at low velocities, where they can exacerbate stick-slip behavior and increase steady-state error. As a result, accurate friction modeling must account for both joint-level and actuator-level contributions to effectively improve control performance.

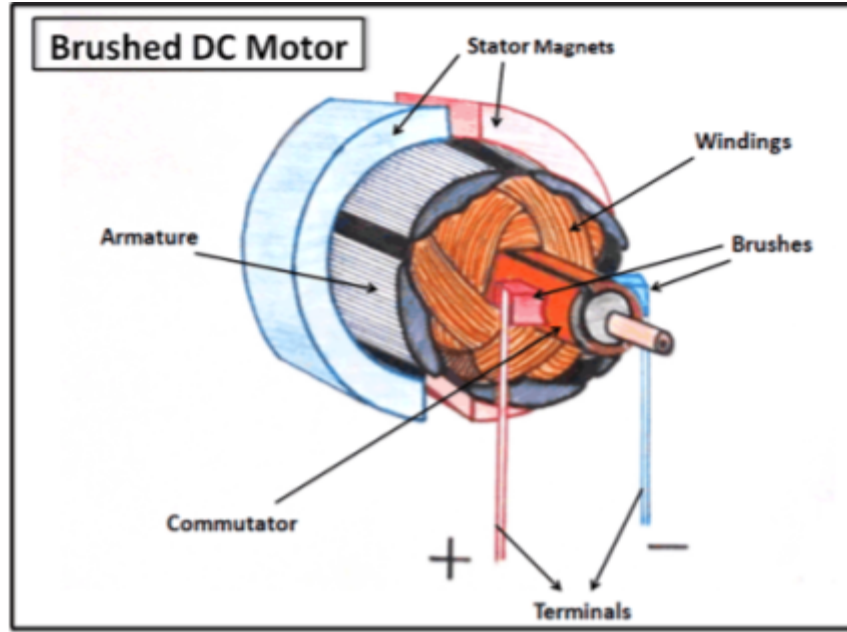


Figure 3: Diagram of a brushed DC motor, demonstrating that contract friction exists on the actuator shaft due to the brushes [4]

In ideal system modeling, no-slip assumptions are often made, implying that motion occurs without relative displacement at contact interfaces. However, in real systems, these assumptions do not hold, and deviations from ideal behavior introduce errors in tracking accuracy, repeatability, and precision control. Friction can lead to phenomena such as steady-state error, stick-slip motion, and degraded performance near zero velocity, making it a critical factor in control design.

Yang et al. (2018) present a friction feedforward sliding mode controller applied to the TriMule hybrid robot for high-precision trajectory tracking in manufacturing applications [5]. Their approach combines a velocity-dependent friction model with a robust sliding mode controller to compensate for nonlinear friction effects. The results demonstrate improved tracking accuracy, particularly for smooth trajectories, highlighting the effectiveness of combining model-based compensation with robust feedback control.

Åström and Canudas de Wit (2008) describe the LuGre friction model, which extends earlier models by incorporating internal dynamic states to represent microscopic contact interactions [6]. This model captures key friction phenomena, including the Stribeck effect and pre-sliding displacement, enabling accurate representation of stick-slip behavior. While the LuGre model provides improved accuracy, it introduces additional complexity in parameter identification and implementation.

Márton and Lantos (2009) further examine friction in conjunction with backlash, using a Stribeck-based friction model to capture velocity-dependent effects under lubricated contact conditions [3]. Their work shows that friction and backlash together can produce complex nonlinear behaviors, including oscillations and limit cycles, emphasizing the importance of modeling both effects within the control framework.

2.3 Control Challenges

The presence of backlash and friction introduces significant challenges in the control of robotic systems, as both nonlinearities lead to discontinuous and non-smooth system behavior. Backlash introduces dead zones and hysteresis, resulting in delayed response and reduced accuracy during direction changes, while friction introduces velocity-dependent disturbances and stick-slip effects, particularly at low speeds. These nonlinearities violate the assumptions of many classical control methods, such as linearity and smooth system dynamics, leading to degraded performance when using standard controllers like PID. As a result, achieving accurate and stable tracking requires control strategies that explicitly model and compensate for these effects, rather than treating them as small disturbances.

3. Approach:

The overall system consists of a model-based control framework that combines data-driven identification of backlash and friction with a feedback controller augmented by feedforward compensation. The approach integrates system modeling, control design, and experimental evaluation to improve tracking performance under varying nonlinear conditions.

3.1 System Modeling

The modeling approach focused on identifying practical representations of backlash and friction for a single-joint robotic system. Several backlash modeling strategies were considered, including hysteresis mapping, two-inertia dynamic modeling, and optimal control approaches such as LQ control. Because the experimental apparatus was designed to support variable and initially unknown backlash levels, a purely assumed backlash gap was insufficient. Instead, backlash was identified directly from encoder data by detecting intervals during which the motor-side incremental encoder changed while the output-side absolute encoder remained approximately constant. These intervals were interpreted as motion through the backlash dead-zone.

For friction, several modeling options were considered, including Coulomb, viscous, and Stribeck friction models. To maintain a tractable model suitable for implementation, the final approach used a Coulomb friction model to estimate the dominant friction coefficient. This coefficient was then used as a feedforward compensation term in the control framework. More complex friction models were considered but not implemented because they added parameter-identification complexity without providing clear additional benefit for the available experimental data.

3.2 Control Strategy

The control strategy began with a baseline PID controller, with gains tuned experimentally through iterative testing. This PID controller was then extended using model-based compensation for the identified nonlinearities (Fig. 4). The friction estimate was incorporated as a feedforward term, while the backlash estimate was used to adjust the joint-position relationship used by the controller. This combined strategy retained the stability and responsiveness of feedback control while adding compensation for repeatable nonlinear effects in the actuator system.

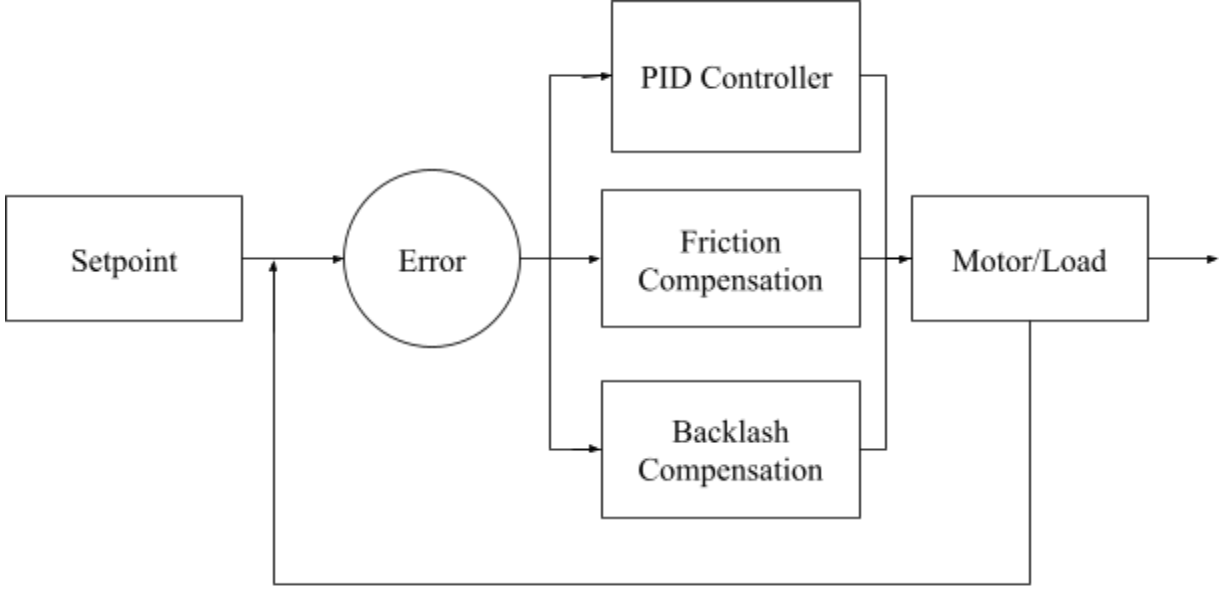


Figure 4. Control block diagram showing PID feedback combined with friction and backlash compensation.

3.3 Evaluation Strategy

The effectiveness of the approach was evaluated by comparing tracking performance between the baseline PID controller and the compensated controller. Encoder measurements were used to quantify system error before and after compensation. A successful controller was defined by a measurable reduction in tracking error across trials with varying levels of backlash and friction.

4. Methods:

4.1 Control Law Implementation

The initial control law implemented for this system was a proportional–integral–derivative (PID) controller. The system dynamics were approximated using a simplified form of the standard manipulator equation:

$$\tau = M(\theta) \ddot{\theta} + C(\theta, \dot{\theta}) \dot{\theta} + g(\theta)$$

For the experimental setup (Section 4.3), the dominant term was the inertia term. The horizontal orientation of the joint minimized gravitational effects, and the system operated at relatively low velocities, making the Coriolis and centrifugal terms negligible. Under these assumptions, the system dynamics were approximated as:

$$\tau \approx M \ddot{\theta}$$

The control input was defined using a PID control law based on tracking error, $\theta_e = \theta_{ref}(t) - \theta(t)$:

$$\tau = K_p \theta_e + K_i \int \theta_e dt + K_d \frac{d}{dt} \theta_e$$

Assuming the control input is proportional to torque, the system can be approximated as:

$$M(\ddot{\theta}) = \tau = K_p e(t) + K_i \int e(t) dt + K_d \frac{d}{dt} e(t)$$

The Coriolis, centrifugal, and gravity terms can be grouped in the end as an h term, and so the full control logic can be defined as:

$$\tau = \ddot{M}(\theta) (\ddot{\theta}_d + K_p \theta_e + K_i \int (\theta_e(t) dt) + K_d \dot{\theta}_e) + \ddot{h}(\theta, \dot{\theta})$$

For this controller, we tuned it iteratively, beginning with values of one for each of the gains (Fig. 5) and refining these parameters until the error range was reduced by approximately 45.5 %, which we determined was significant error reduction for the tuned parameters to be sufficient. The final gains for our PID controller were $K_p = 1.5$, $K_i = 0.5$, and $K_d = 0.05$ (Fig. 6)

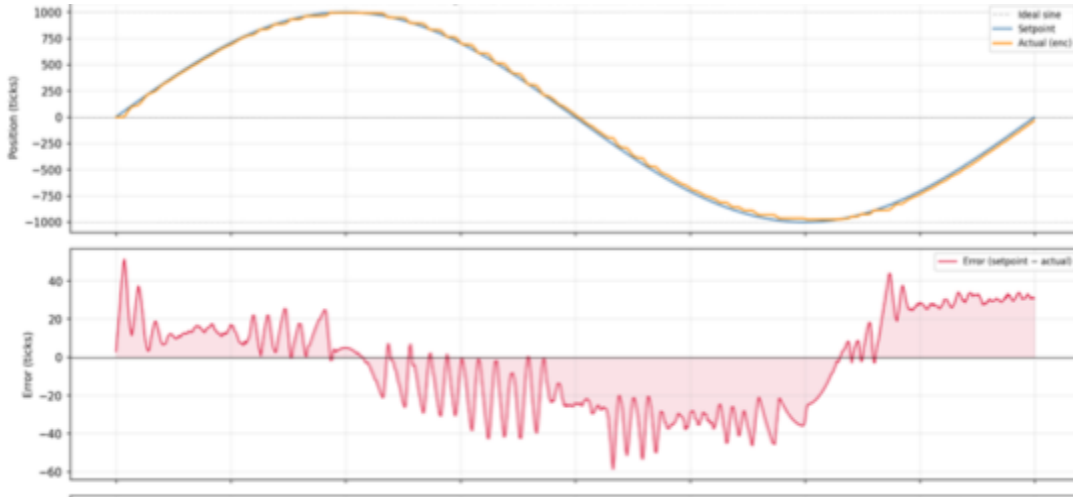


Figure 5: Setpoint trajectory vs actual trajectory from encoders for $K_p = K_i = K_d = 1.00$ with error range of 110 ticks.

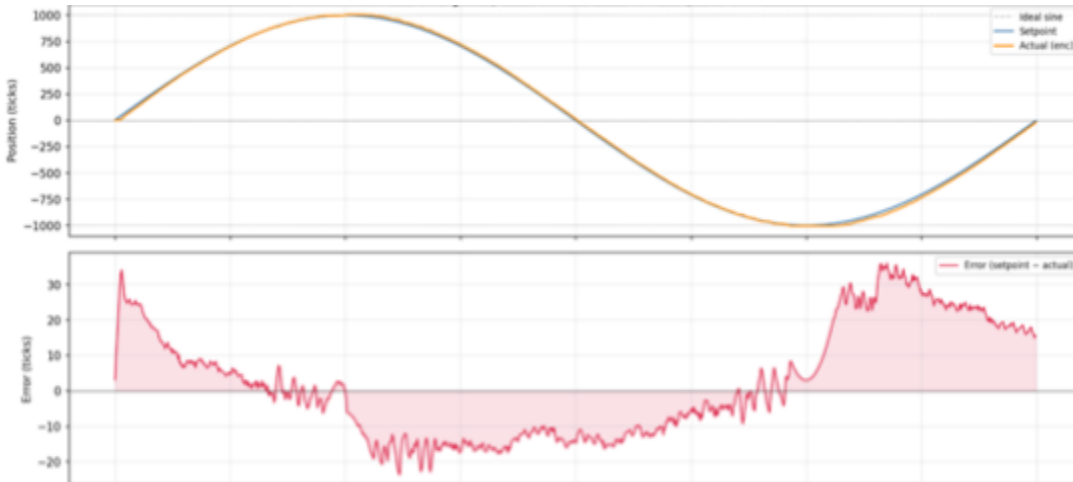


Figure 6: Setpoint trajectory vs actual trajectory from encoders for $K_p = 1.5$, $K_i = 0.5$, $K_d = 0.05$ with error range of 60 ticks.

4.2 Model Formulation

An initial MATLAB model was developed to evaluate how backlash and friction could be represented in a single-joint actuator system prior to experimental testing. In this model, the motor-side dynamics were simulated as a one-degree-of-freedom rotational system, where the applied torque was opposed by friction and load torque:

$$J_{eq} \ddot{\theta}_m = \tau_{cmd} - \tau_f(\dot{\theta}_m) - \tau_{load}$$

where J_{eq} is the equivalent inertia, θ_m is the motor-side angle, τ_{cmd} is the commanded torque, and τ_f is the modeled friction torque.

Friction was initially implemented using a progressive modeling framework, beginning with Coulomb friction and allowing additional terms such as viscous friction and Stribeck behavior to be incorporated if needed. The most general friction model considered was:

$$\tau_f(\dot{\theta}) = (F_d + (F_s - F_d)e^{-(|\dot{\theta}|/v_s)^2})\tanh(\frac{\dot{\theta}}{\epsilon}) + B\dot{\theta}$$

Backlash was represented using a hysteretic dead-zone mapping between the motor-side angle and output angle, as described in Section 4.2.1.

This initial simulation framework was used to generate synthetic torque, position, and velocity data and to compare candidate friction models before experimental data were collected. The simulation was useful for validating the overall identification workflow, but the final experimental implementation was simplified to use Coulomb friction and data-driven backlash identification.

FIGURE

4.2.1 Backlash Model

The initial backlash model used a dead-zone (hysteresis) mapping between motor-side and output-side position:

$$\theta_o = \begin{cases} \theta_m - b & \text{if } \theta_m - \theta_o > b \\ \theta_m + b & \text{if } \theta_m - \theta_o < -b \\ \theta_o & \text{if } |\theta_m - \theta_o| \leq b \end{cases}$$

where b is the backlash half-width.

This approach assumed a known backlash parameter and predicted output motion accordingly. However, in this system the backlash was intentionally varied and not known a priori, making this formulation

unsuitable. The resulting model did not accurately capture the observed system behavior, and instead showed a consistently high value of backlash in the system with no variation (Fig. 7).

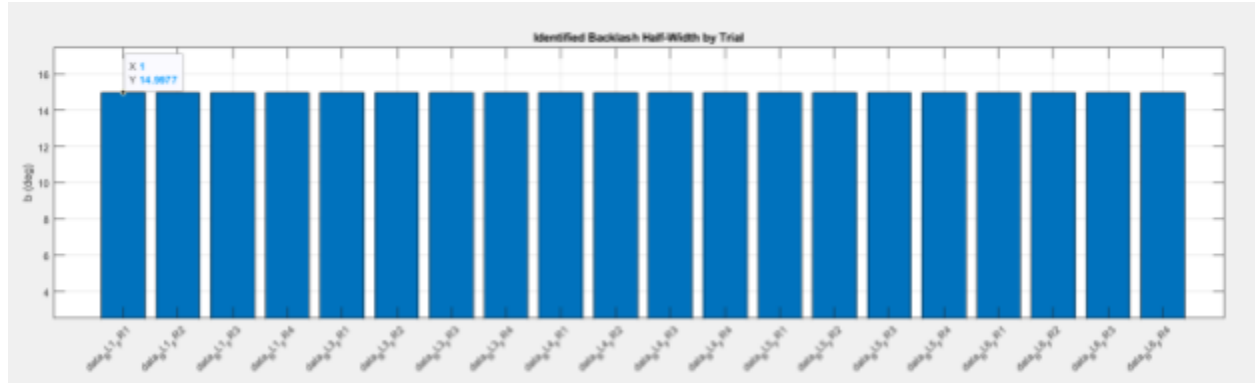


Figure 7: Backlash half-width across all trials using hysteresis/deadzone mapping

To address this limitation, the model was redefined using an experimental identification approach. Backlash was determined directly from encoder data by identifying regions where the output position remained approximately constant while the motor position continued to change. The backlash magnitude was computed as:

$$b = \frac{\Delta\theta_m}{2}$$

where $\Delta\theta_m$ is the motor displacement during the dead-zone region.

This data-driven approach produced results consistent with the expected increase in backlash across experimental conditions (Fig. 8).

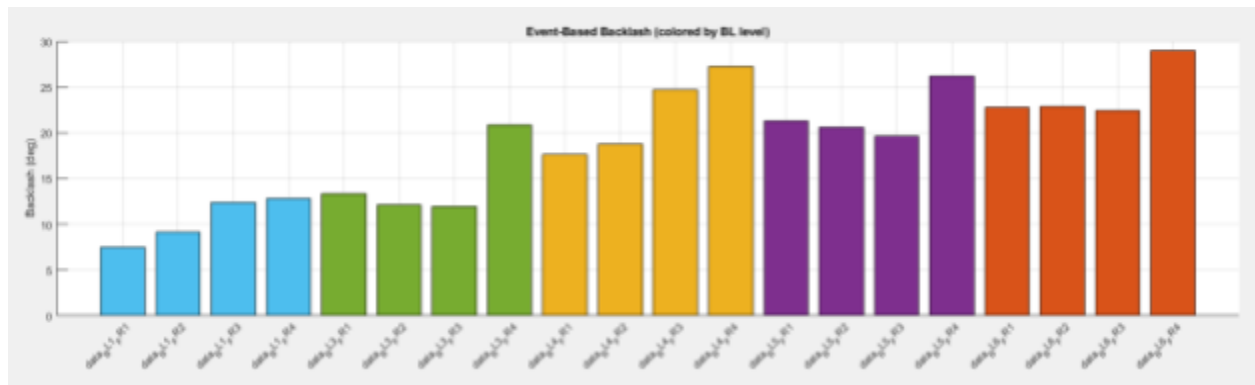


Figure 8: Backlash in each of the trials by experimental data

4.2.2 Friction Model

For the final model, friction was represented using a Coulomb friction formulation, as it captured the dominant nonlinear behavior observed in the system while maintaining simplicity:

$$\tau_f = F_c \text{sign}(\dot{\theta})$$

To improve numerical stability and avoid discontinuities near zero velocity, a smoothed formulation was used:

$$\tau_f = F_c \tanh\left(\frac{\dot{\theta}}{\epsilon}\right)$$

Initial results showed discrepancies between predicted and measured friction behavior. Upon further investigation, it was determined that the model did not correctly account for the direction of motion, leading to incorrect cancellation of friction forces.

This was corrected by ensuring that friction was consistently applied as an opposing force to motion, preserving its sign dependence. With this refinement, the model produced friction trends that matched the experimental data.

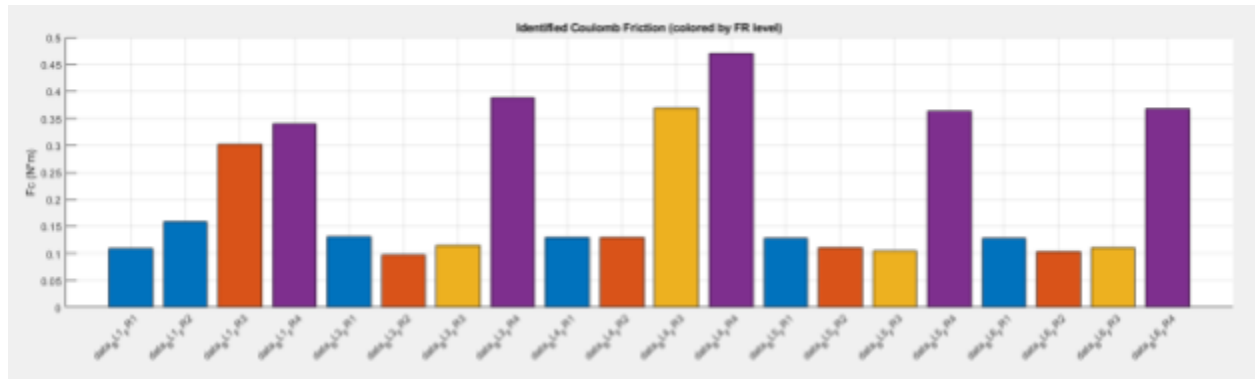


Figure. 9. Coulomb friction coefficients across all trials. Friction levels are color-coded (blue: lowest, orange: second, yellow: third, purple: highest).

4.3 Apparatus Development

The experimental apparatus consisted of a 3D-printed PLA motor mount (Fig. 10), along with modular components designed to introduce controlled levels of friction and backlash into the system. The actuator used was a 12V DC gearmotor (251 RPM) equipped with an incremental encoder for motor-side measurements. Load-side position was measured using a single-turn SSI absolute encoder.

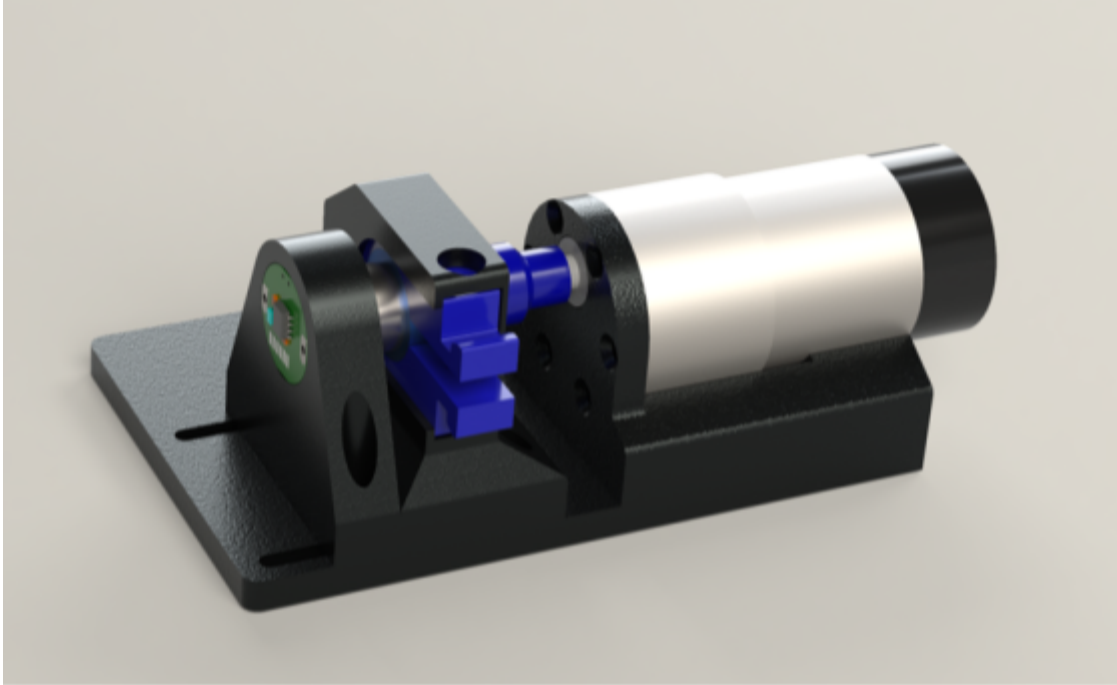


Figure 10: Solidworks model of the experimental apparatus

Friction was introduced using a clamp mechanism that fully surrounded the rotating shaft. The friction level was adjusted by tightening a screw, which increased the normal force applied to the shaft. Four discrete levels of friction were defined based on the position of the screw, with the screw divided into thirds and tightened in increments of two-thirds of a turn for each level (Fig. 11).



Figure 11: (Left) Friction inducer, the separated distance (red) was how we numerically assigned the friction level; (Right) The screw that increased the friction, with the thirds that we followed marked in red.

Backlash was introduced using interchangeable coupling components between the motor output shaft and the load-side shaft. The coupling consisted of a D-shaft insert to interface with the motor and a mating geometry connected to the backlash inducer. The backlash inducers were designed with rounded internal corners, and increasing the radius reduction resulted in greater mechanical play between mating components. Six discrete levels of backlash were implemented by varying this geometry (Fig. 12).

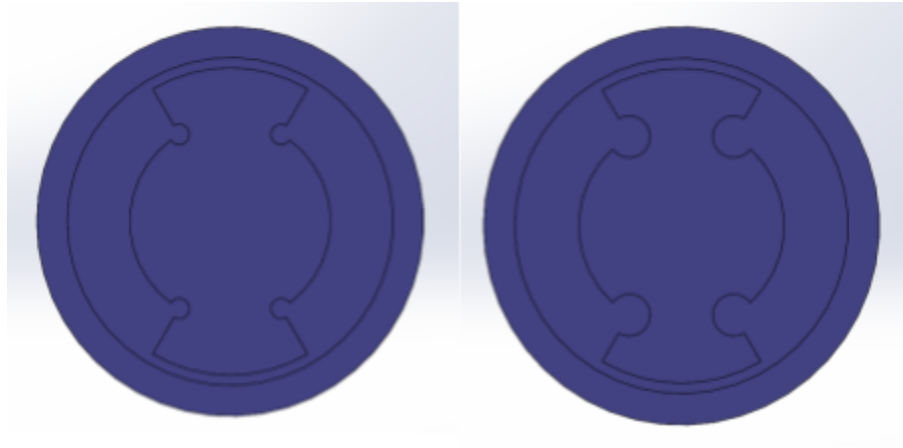


Figure 12: (Left) Backlash inducer with largest gap, (Right) Backlash inducer with smallest gap



Figure 13: Backlash inducers for all levels; (Leftmost) Shaft output

The electrical system was built around an Arduino Uno microcontroller and a Pololu Dual VN5019 motor driver shield. An initial motor driver (TB6612FNG) was found to be insufficient due to current limitations under high-friction conditions, and was replaced with the VN5019 to accommodate higher current demands. The SSI absolute encoder communicated with the microcontroller using a synchronous serial interface, in which a clock signal generated by the Arduino was used to read position data bit-by-bit. The motor driver also provided real-time current measurements, which were logged and later used for friction model identification (Section 4.4).

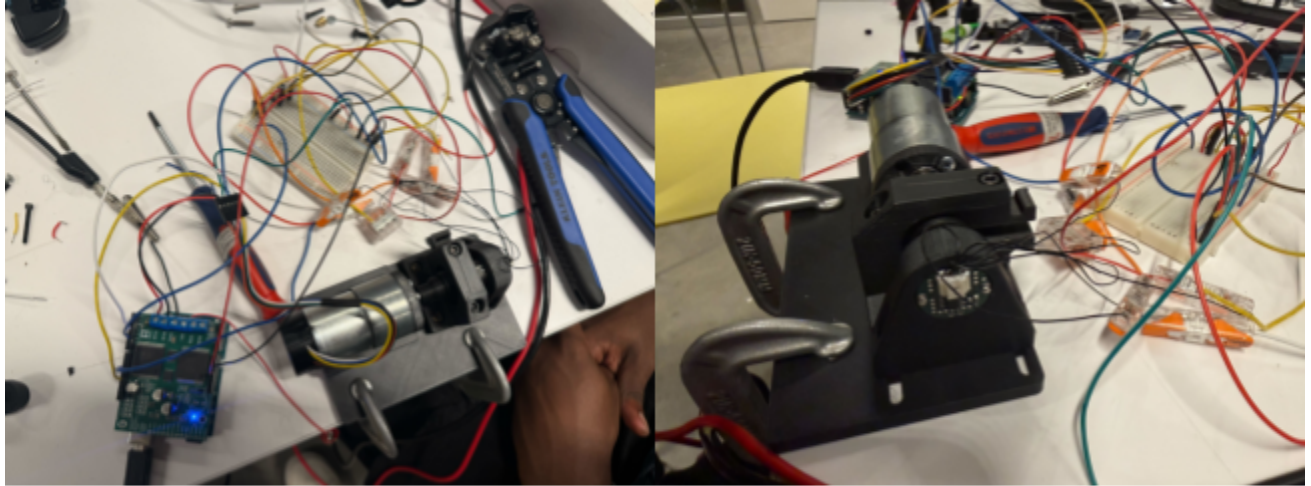


Figure 14: Setup Experimental Apparatus

4.4 Data Collection and Parameter Identification

To evaluate the system, a total of 20 experimental trials were conducted (Table 1), corresponding to all combinations of the six backlash levels and four friction levels implemented in the apparatus. For each trial, the system was operated using the baseline PID controller, and time-series data were recorded throughout the motion (Fig. 15).

Table 1: Experimental Trials						
		Backlash Levels				
	-	Default	0.1 mm	0.3 mm	0.4 mm	0.5 mm
Friction Levels	0.212 in	Trial 01	Trial 02	Trial 03	Trial 04	Trial 05
	0.205 in	Trial 06	Trial 07	Trial 08	Trial 09	Trial 10
	0.195 in	Trial 11	Trial 12	Trial 13	Trial 14	Trial 15
	0.162 in	Trial 16	Trial 17	Trial 18	Trial 19	Trial 20

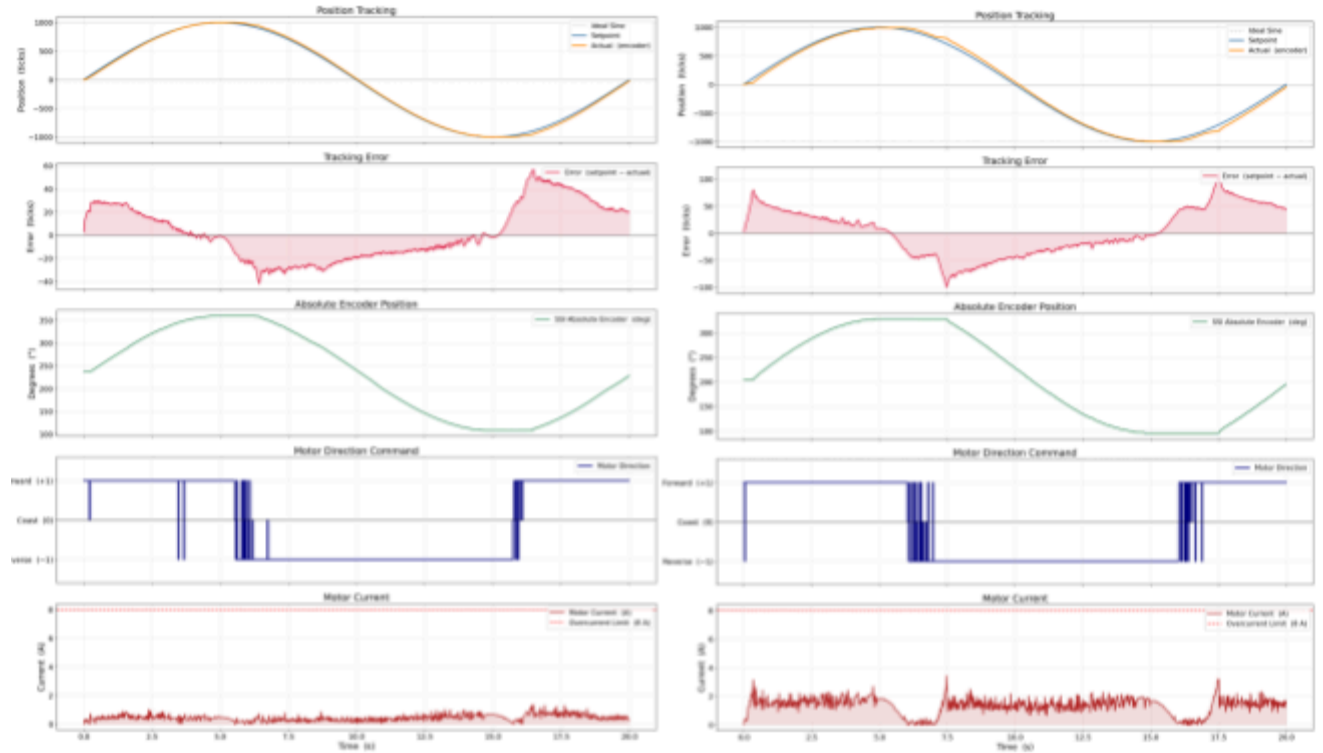


Figure 15: (Left) Time series data for lowest levels of backlash and friction, (Right) Time series data for highest levels of backlash and friction

The following variables were collected at each timestep:

- time (ms and s)
- reference position (setpoint)
- measured position (incremental encoder)
- PWM command
- motor direction
- system state
- absolute encoder count and angle (SSI)
- motor current

From these measurements, additional variables were computed, including tracking error and signed current. Each trial produced approximately 995 data points, which were stored as CSV files for further analysis.

The collected data were processed using the previously developed friction and backlash identification models. Friction parameters were estimated by relating motor current to torque and fitting a Coulomb friction model to velocity-dependent data. Backlash was identified using an event-based approach, where regions of constant output position (from the absolute encoder) during motor motion were used to estimate the backlash gap.

For each trial, the following parameters and metrics were extracted:

- friction coefficient and smoothing parameter
- friction model error (RMSE)
- mean and absolute current values
- backlash magnitude (in motor counts, degrees, and radians)
- standard deviation of backlash measurements
- number of detected backlash events
- number of samples used for fitting
- average timestep

These values were compiled into a summary table (Table 2) for comparison across trials.

Table 2: Parameters Identified in Model		
Backlash Level	Avg Backlash (deg)	Friction Range
BL1	~10	0.11 – 0.34
BL3	~14	0.10 – 0.39
BL4	~22	0.13 – 0.47
BL5	~22	0.10 – 0.36
BL6	~24	0.10 – 0.37

The identified friction and backlash parameters were then incorporated into the updated control law. Specifically, the friction coefficient was used to generate a feedforward compensation term, while the backlash estimate was used to adjust the effective position relationship in the controller. The performance of this compensated controller was then evaluated against the baseline PID controller.

5. Results:

Time-series data from the compensated controller were compared against the baseline PID controller across all tested friction and backlash conditions (Fig. 16). For the lowest friction and backlash case, the compensated controller reduced the tracking error range from approximately 60 ticks to 30 ticks, corresponding to a 50% reduction in error. This demonstrates a clear improvement in tracking accuracy when incorporating model-based compensation.

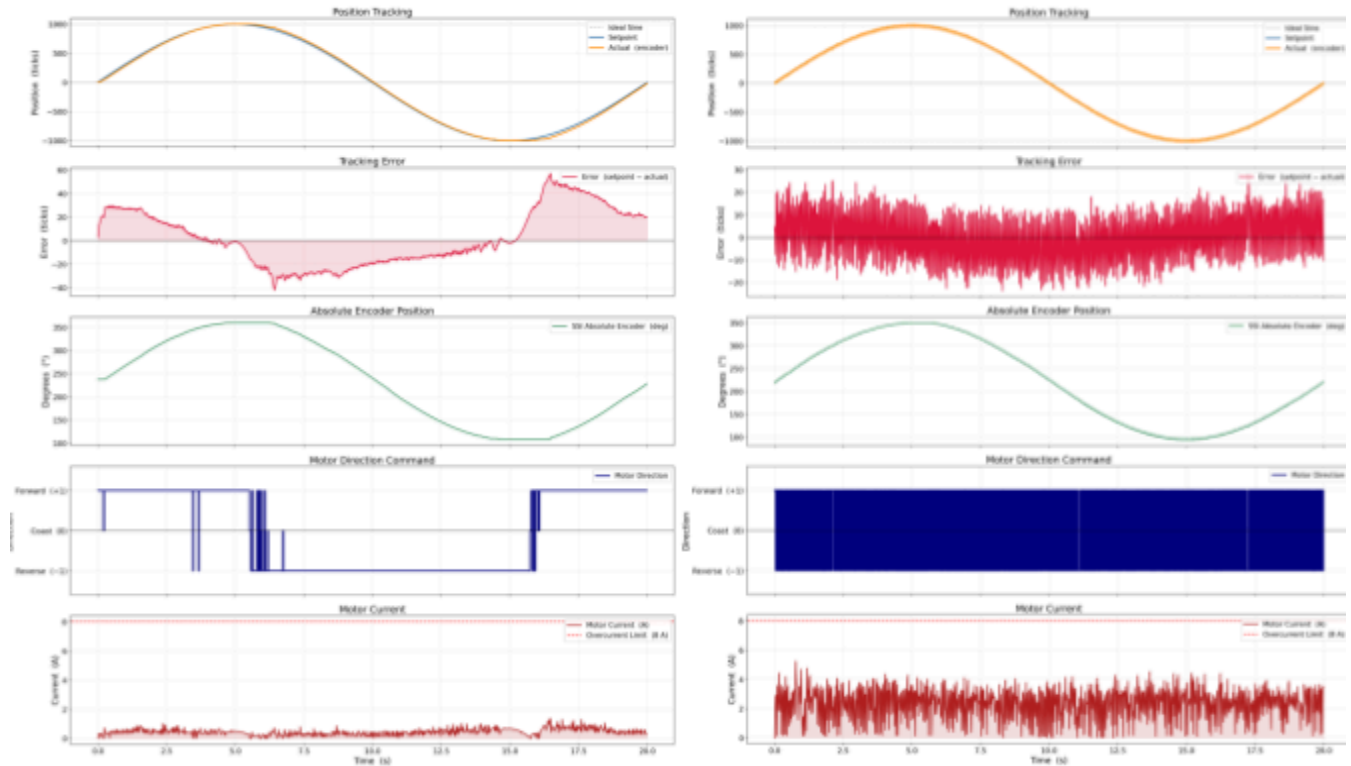


Figure 16: PID Controller vs Friction/Backlash Compensated Controller for Lowest levels of friction and backlash

As the levels of friction and backlash increased, the overall tracking error also increased, as observed in the baseline PID-only trials. However, the compensated controller consistently reduced the magnitude of this error across the tested conditions, indicating that the proposed approach remains effective even as system nonlinearities become more pronounced.

Despite these improvements, the inclusion of friction and backlash compensation introduced additional control effort. In particular, the controller produced more frequent direction changes and higher torque demands, which resulted in significant increases in motor current. These current surges were especially noticeable near direction reversals, where the controller actively compensates for backlash and friction effects.

At higher friction and backlash levels, the increased current demand exceeded the safety limits implemented in the system. A current cutoff was enforced in software to prevent damage to the motor driver, which had previously failed under high-load conditions. As a result, the compensated controller could not be fully evaluated for the most extreme test cases.

6. Takeaways:

This project demonstrates that explicitly modeling and compensating for friction and backlash can significantly improve the performance of real robotic systems compared to relying on idealized kinematic assumptions. The experimental results show that even a relatively simple model using Coulomb friction

and an empirically identified backlash dead-zone can effectively capture the dominant nonlinear behaviors affecting a single-joint actuator. Incorporating these effects into the control law led to a substantial reduction in tracking error, particularly in low-friction and low-backlash conditions, where the error range was reduced by approximately 50%. More importantly, the approach remained effective as system nonlinearities increased, consistently improving performance relative to a baseline PID controller. The event-based backlash identification method proved to be both practical and interpretable, allowing direct extraction of meaningful system parameters from encoder data without requiring complex modeling assumptions. Overall, this work highlights that bridging the gap between ideal models and real-world behavior does not necessarily require highly complex formulations; rather, targeted modeling of dominant nonlinearities can yield significant improvements in accuracy and system understanding.

7. Future Considerations:

While the proposed framework improved tracking performance, it also revealed important limitations that motivate future work. One key area for improvement is the definition and modeling of friction within the system. In this work, friction was introduced and characterized using a mechanical clamp with discrete levels defined by screw position, which may not provide a consistent or repeatable measure of applied friction. Future iterations of the apparatus could incorporate more precise and quantifiable methods, such as calibrated force application or the use of pin gauges, to better control and characterize friction levels. Additionally, while a Coulomb friction model captured the dominant behavior, more advanced models (viscous, Stribeck, etc.) could be revisited to improve accuracy, particularly in low-velocity regimes.

Another important consideration is the increased control effort required by the compensated controller. The addition of friction and backlash compensation led to higher torque demands and frequent direction reversals, resulting in significant current surges. These effects limited the ability to test the controller under high-friction and high-backlash conditions due to enforced current limits designed to protect the hardware. Future work should therefore focus on refining the control strategy to balance performance improvements with actuator constraints. This could include incorporating current-aware control limits, smoothing or filtering compensation terms to reduce abrupt control actions, or integrating the actuator dynamics and current limitations directly into the model. More broadly, extending the model to account for electrical and thermal constraints of the motor driver would enable a more realistic and robust control framework, ensuring that performance gains do not come at the cost of system reliability.

Another important direction for future work is the independent evaluation of friction and backlash compensation through decoupled control strategies. In the current implementation, both nonlinearities were addressed simultaneously within a single augmented controller, making it difficult to isolate the individual contribution of each compensation method to overall performance improvement. Future studies could implement separate control configurations, such as one incorporating only friction feedforward compensation and another incorporating only backlash compensation, while keeping all other control parameters constant. This would allow for a more systematic comparison of how each nonlinearity affects tracking accuracy, stability, and control effort. Additionally, such an approach could reveal interactions between the two compensation terms, including whether they introduce conflicting control actions under certain operating conditions. By isolating and evaluating each component independently, the effectiveness and limitations of each model can be more clearly understood, ultimately informing the design of more optimized and balanced control strategies.

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Appendix:

Appendix A: Bill Of Materials

Bill of Materials		
Component Description	Supplier	Part Number
Absolute Encoder (AMT112S, 4096 CPR)	Digi-Key	2223-AMT112S-4096-5000-S-ND
SSI Absolute Encoder (SparkFun)	Digi-Key	1568-20359-ND
DC Gearmotor (12V, 251 RPM)	Digi-Key	1738-1106-ND
Motor Driver Shield (Pololu Dual VNH5019)	Pololu	2507
Arduino Uno Microcontroller	Arduino	A000066